a 2-dimensional control space given by its acceleration.

The environment considered is shown in Figure 2. There are ℓ landmarks placed in the environment, which are represented by black stars. The black polygonal shapes represent the obstacles. The agent observes all landmarks and obtains estimates of its position at all time steps. The agent achieves better position estimates when it is closer to the landmarks. Let the Euclidean distance between the position of the 2-D double integrator and the jth landmarks be given by d_j. Then, the jth position measurement corresponding to landmark j is

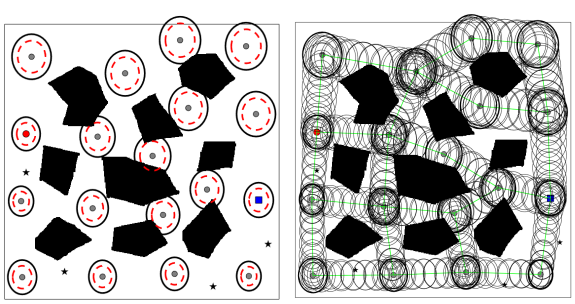
$$^{j}y = [x^{(1)}\ x^{(2)}]^\top + \eta_p d_j v_p, \quad j = 1, 2, \cdots, \ell, \tag{24}$$

where η_p is a parameter related to the intensity of the position measurement noise that is set to 0.1, and v_p is a two-dimensional standard Gaussian random vector. The velocity measurement is given by

$$y_v = [x^{(3)}\ x^{(4)}]^\top + \eta_v v_v, \tag{25}$$

where η_v is a parameter related to the intensity of the velocity measurement noise that is set to 0.2, and v_v is a two-dimensional standard Gaussian random vector. Thus, the total measurement vector y is a $2\ell+2$ dimensional vector, composed of ℓ position measurements and one velocity measurement.

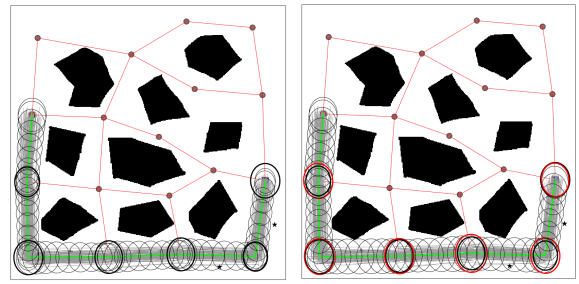
The sampled CS-BRM nodes are shown in Figure 2(a). The velocities are restricted to be zero for the purpose of comparison with SLQG-FIRM. The gray dots are the state means of the nodes. The red dot shows the position of the starting node and the blue square shows the position of the goal node. The black ellipses around the state represent the 3σ confidence intervals of the position covariances. The red dash ellipses correspond to the prior estimation error covariances. Following Algorithm 2, the edge controllers are computed using the covariance steering controller for each edge and the collision cost is computed using Monte Carlo simulations. The final CS-BRM is given in Figure 2(b).



(a) (b)

Fig. 2: (a) Sampled CS-BRM nodes. (b) CS-BRM of the 2-D double integrator. The green lines are the mean trajectories between the nodes. The gray ellipses are the 3σ confidence intervals of the covariances of the positions.

After the CS-BRM is built, the path planning problem on CS-BRM is the same as the problem of planning using a PRM, which can be easily solved using a graph search algorithm. The difference between the CS-BRM and PRM is that the edge cost in CS-BRM is specifically designed to deal with dynamical system models and uncertainties, and the transition between two nodes of CS-BRM is achieved using covariance steering as the edge controller.



(a) (b)

Fig. 3: Results of sampling stationary belief nodes. (a) Path planned using CS-BRM; (b) Path planned using SLQG-FIRM.

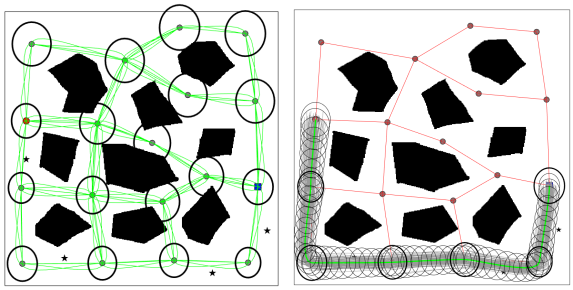
The planning results of CS-BRM and SLQG-FIRM are given in Figure 3. The paths goes through the same set of belief nodes for this example. In Figure 3(b), the red ellipses are the state covariances at the last step of the time-varying LQG controller. As we see, each one of the ellipses is larger than the corresponding state covariance of the intermediate belief node (bold black ellipses). Thus, a converging step is required at every intermediate node. The time-varying LQG controller is switched to the SLQG controller until the state covariance converges to a value that is smaller than the state covariance of the belief nodes (bold black ellipses).

The CS-BRM that samples nonstationary nodes is shown in Figure 4(a). For clarity, only the mean trajectories of the edges are shown (green lines). The environment, the sampled mean positions, and the sampled covariances are all the same as those in Figure 3(a) and 3(b). The only difference is that multiple velocities are sampled at each position in Figure 4(a). The planned path is shown in Figure 4(b). The cost of the planned path is 104.87, which is much lower than the cost of the paths in Figure 3(a) and Figure 3(b), which are 239.36 and 240.47 respectively. The decrease in the path cost is due to the decrease of the mean control cost. In Figure 3(b), the robot has to stop (zero velocity) at every intermediate node. With the proposed method, the robot goes through each intermediate node smoothly, which results in a more efficient path.

B. Fixed-Wing Aircraft

In this section, we apply the proposed method on a nonlinear example, namely, planning for a fixed-wing aerial vehicle [32]. The discrete-time system model is given by

$$\begin{aligned} x_{k+1} &= x_k + V_k \cos\psi_k \cos\gamma_k \Delta t + g_{1k} w_{1k}, \\ y_{k+1} &= y_k + V_k \sin\psi_k \cos\gamma_k \Delta t + g_{2k} w_{2k}, \\ z_{k+1} &= z_k + V_k \sin\gamma_k \Delta t + g_{3k} w_{3k}, \\ \psi_{k+1} &= \psi_k + \frac{g}{V_k} \tan\phi_k \Delta t + g_{4k} w_{4k}, \end{aligned} \tag{26}$$



(a) (b)

Fig. 4: (a) CS-BRM results of sampling nonstationary nodes. The green lines are the mean trajectories of the edges. (b) Planning results. By exploring the velocity space, the robot does not need to stop at every node. The path cost is much smaller compared to Figures 3(a) and 3(b).

where the 4-dimensional state space is given by $[x\ y\ z\ \psi]^\top$, where $[x\ y\ z]^\top$ is the 3-D position of the vehicle, and ψ is the heading angle. The 3-D control input space is given by $[V\ \gamma\ \phi]^\top$, where V is the air speed, γ is the flight-path angle, and ϕ is the bank angle, Δt is the time step size, w_{ik}, $i = 1, 2, 3, 4$, are standard Gaussian random variables, g_{1k}, g_{2k}, g_{3k}, and g_{4k} are multipliers correspond to the magnitude of the noise. Their values are all set to be 0.1.

Similar to the 2-D double integrator example, several landmarks are placed in the environment. Let the Euclidean distance between the 3D position of the vehicle and the jth landmarks be given by d_j. Then, the jth position measurement corresponding to landmark j is

$$^{j}y = [x\ y\ z\ \phi]^\top + \eta d_j v, \quad j = 1, 2, \ldots, \ell, \tag{27}$$

where η is a parameter related to the intensity of the noise of the measurement and is set to 0.05, and v is a 4-dimensional standard Gaussian random vector. Thus, the total measurement vector y is a 4ℓ-dimensional vector, where ℓ is the number of landmarks.

Algorithm 1 was used to find a compatible nominal trajectory for the fixed-wing vehicle. One example is given in Figure 5. The nominal trajectory and the reference trajectory are initialized as straight lines in all four state dimensions and the three control input dimensions. The algorithm converged in eight iterations.

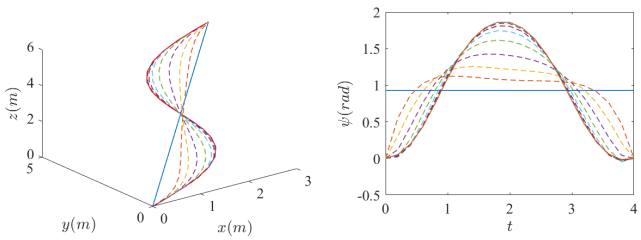


Fig. 5: Generation of a compatible nominal trajectory for the fixed-wing vehicle. The nominal trajectory and the reference trajectory are initialized as straight lines.

The planning environment is shown in Figure 6. The four spheres in the middle of the environment are the obstacles. The three black stars represent the landmarks. The black circles represent the state mean (position) of the nodes. The red dot shows the position of the starting node and the blue square shows the position of the goal node. The state means are deterministically chosen to discretize the environment. The covariances of the nodes are not showed.

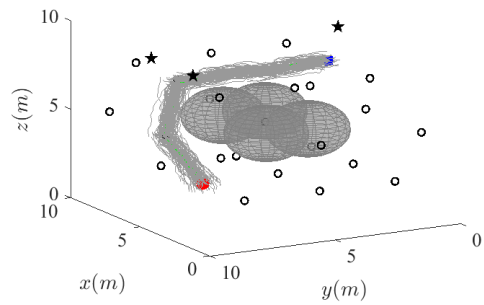


Fig. 6: Planning results using the CS-BRM for the fixed-wing vehicle. The gray lines are the trajectories from the Monte-Carlo simulations. The algorithm finds the optimal path that trades off between control effort and collision cost.

The planned path using this CS-BRM is also shown in Figure 6. Similarly to the example in Section VI-A, instead of flying through the narrow passage between the four obstacles, which resulting a high probability of collision, the algorithm choose a path that trades off between control cost and collision cost while minimizing the total edge costs.

VII. Conclusion

A belief space roadmap (BRM) algorithm is developed in this paper. The nodes in BRM represent distributions of the state of the system and are sampled in the belief space. The main idea is to use covariance steering to design the edge controllers of the BRM graph to steer the system from one distribution to another. Compared to [14], the proposed method allows sampling nonstationary belief nodes, which has the advantage of more complete exploration of the belief space. For covariance steering of nonlinear systems, we introduce the concept of compatible nominal trajectories, which aim to better approximate the nonlinear dynamics through successive linearization. We also propose an efficient algorithm to compute compatible nominal trajectories. Compared to the standard PRM, the additional computation load comes from the computation of the edge controllers and edge cost evaluations, which, however, are done offline. By explicitly incorporating motion and observation uncertainties, we show that the proposed CS-BRM algorithm generates efficient motion plans that take into account both the control effort and collision probability.

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